MACRO THEORY · PAPER PRESENTATION

Secular stagnation? The effect of aging on economic growth in the age of automation

Daron Acemoglu & Pascual Restrepo · *AER: Papers & Proceedings* 107(5), 2017

Theory · replication · an extension with PWT 11.0 and UN WPP 2024

THE THESIS

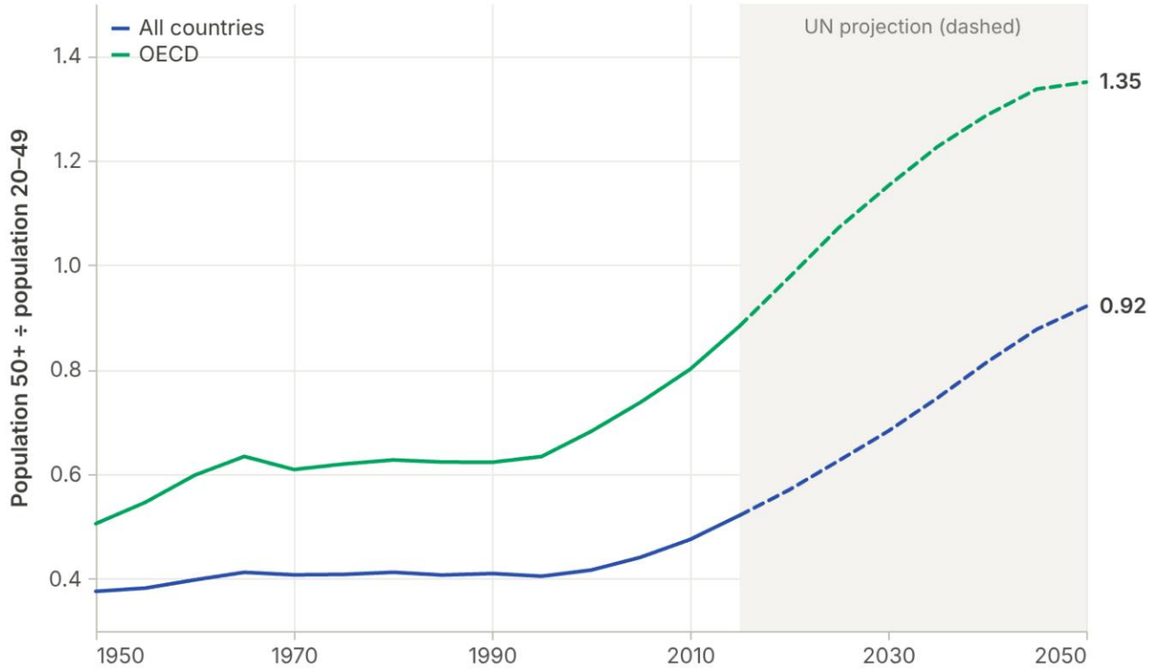
Aging need not mean stagnation

The worry. Hansen (1939), Summers (2013) and Gordon (2016): an older population means excess saving, fewer workers and slower growth, so faster-aging countries should grow more slowly.

Acemoglu & Restrepo's claim. Since 1990 there is **no negative relationship** between aging and GDP per capita growth; if anything, it is positive.

Why? Labor scarcity triggers **automation**. Capital-rich economies gain the most from each task they hand to machines, because capital is cheap relative to labor. So a shrinking workforce pushes them further into automation, which can offset, or even reverse, the drag from fewer workers.

Aging, 1950–2015, and UN projections to 2050
Unweighted mean across the paper's 169 countries (OECD: 35)



Overlay the UN 2024 revision ☐

Figure 1 in the paper, rebuilt from the replication data (UN WPP 2015). Tick the box to overlay the 2024 UN revision: the forecast has barely moved.

Who produces what

An intermediate good is an input that is used up in making something else. Here every task needs one, and a single firm supplies them all.

The monopolist

Sole supplier of the task-specific intermediates $q(i)$. Produces them at marginal cost 1, sells at a markup. **Also chooses θ**: which tasks can be automated.

→

Competitive task producers

Produce each task $i \in [0, 1]$ from $q(i)$ plus capital or labor, taking prices as given. Zero profits.

→

Final good Y

Cobb–Douglas aggregate of all tasks: $\ln Y = \int_0^1 \ln y(i)\, di$

Analogy: one tool company, many builders. Picture a single company that makes the specialized tool every step of building a house needs: a nail gun for framing, a saw for cutting, and so on. Thousands of competing contractors buy those tools and build the houses (the final good). The tool company charges more than its cost (its markup), and it also decides, through R&D, **how many steps its tools can do on their own** without a worker. That share of steps is θ. Automating more steps costs the tool company more, and harder steps cost disproportionately more.

The monopolist's two choices, the price of $q(i)$ and the automation frontier θ, drive everything that follows.

Two kinds of tasks, one essential input

Production is a Cobb–Douglas aggregate over a continuum of tasks: ln Y = ∫₀¹ ln y(i) di.

(2) AUTOMATED TASKS, I ≤ Θ

$$y(i) = \underbrace{q(i)^{\eta}}_{\text{intermediate}} \left(k(i) + l(i)\right)^{1-\eta}$$

Capital and labor are **perfect substitutes**: a machine can do the task.

(3) NON-AUTOMATED TASKS, I > Θ

$$y(i) = \underbrace{q(i)^{\eta}}_{\text{intermediate}} l(i)^{1-\eta}$$

Only labor can do the task.

Both technologies use the intermediate *q(i)*, with the same weight η. That is how the monopolist earns a share of all output, whichever factor does the task. Drag θ to move the line between machine-capable and labor-only tasks.



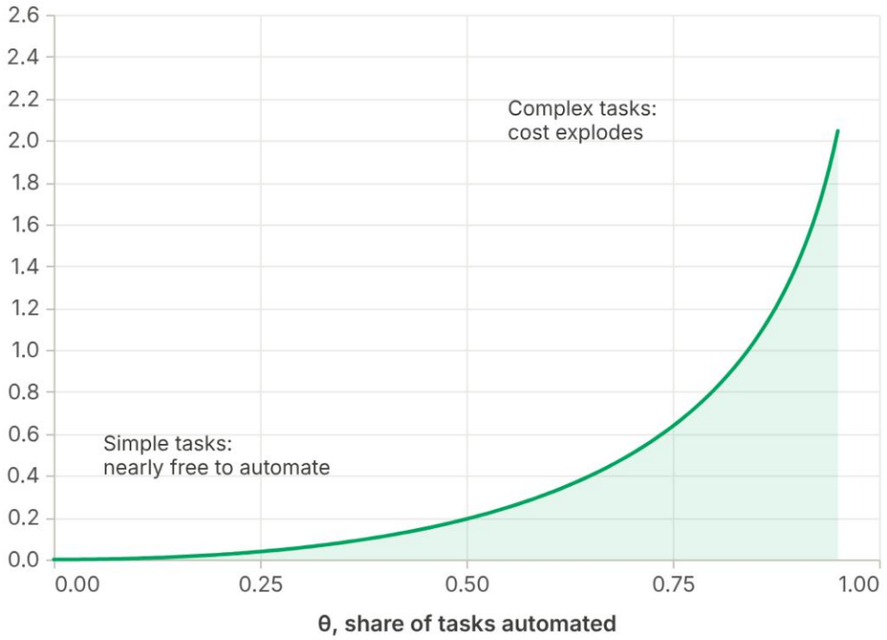
Automation frontier θ 0.6

Automation is cheap at first, then prohibitive

The monopolist pays C(θ)·Y to make tasks 0…θ automatable. Technological difficulty lives entirely in C(·).

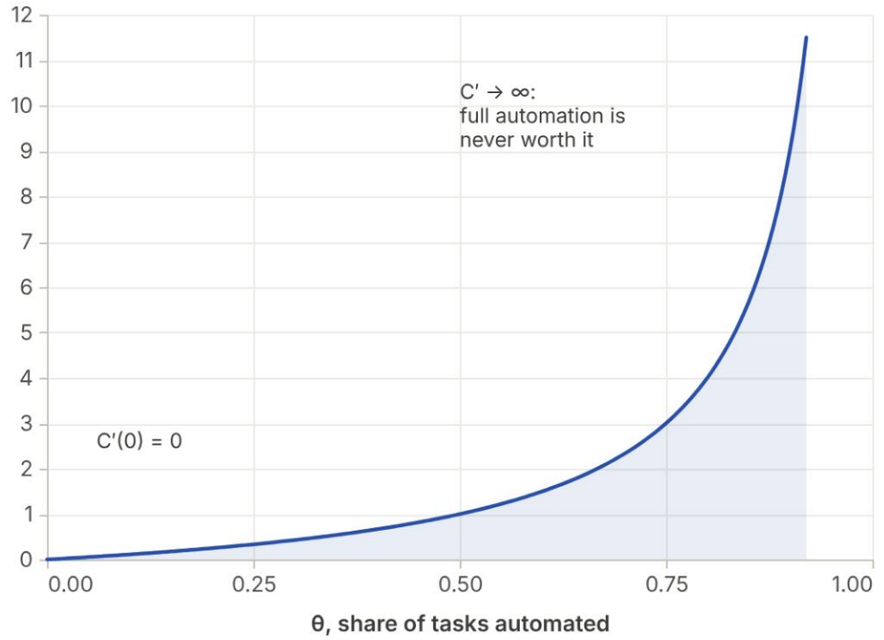
Cost of automating tasks 0…θ: C(θ)

Increasing and convex; finite at every θ < 1



Marginal cost of one more task: C′(θ)

C′(0) = 0 and C′(θ) → ∞ as θ → 1



Illustrative cost function C(θ) = −θ − ln(1 − θ), which satisfies the paper's assumptions (twice differentiable, strictly increasing, strictly convex, Inada conditions on C′). The paper does not choose a functional form. Hover for values.

Equilibrium output and the key condition

EQUILIBRIUM AGGREGATE OUTPUT

$$Y = \eta^{\frac{\eta}{1-\eta}} \left(\frac{K}{\theta}\right)^{\theta} \left(\frac{L}{1-\theta}\right)^{1-\theta}$$

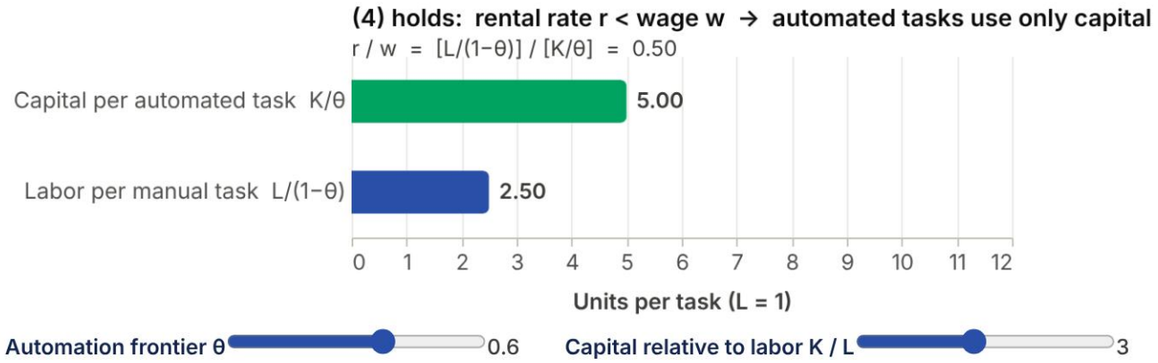
Capital is spread evenly over the θ automated tasks, labor over the 1 – θ manual ones.

CONDITION (4): CAPITAL IS ABUNDANT AND CHEAP RELATIVE TO LABOR

$$\frac{K}{\theta} > \frac{L}{1-\theta} \quad \Longleftrightarrow \quad r < w$$

Why (4) is just "r < w". With Cobb–Douglas tasks, factor prices are inversely proportional to how much of the factor each task gets: $r \propto \theta/K$ and $w \propto (1-\theta)/L$. So $r < w$ exactly when capital per automated task exceeds labor per manual task.

When (4) holds, machines are the cheaper way to do every automatable task, so automating one more task **raises output**: $\partial \ln Y / \partial \theta = \ln \frac{K}{\theta} - \ln \frac{L}{1-\theta} > 0$.



The paper writes the constant as η^(η/(1−η)), which implicitly sets the intermediate's price p = 1. With the markup it is (η/p)^(η/(1−η)). The constant drops out of every derivative, so nothing below depends on it.

The monopolist chooses how much to automate

THE MONOPOLIST'S PROBLEM (IN LOGS)

$$\max_{\theta\in[0,1]} \underbrace{\theta \ln \frac{K}{\theta} + (1-\theta)\ln\frac{L}{1-\theta}}_{\ln Y(\theta)\text{: size of the pie}} + \underbrace{\ln\left(\eta\chi - C(\theta)\right)}_{\Gamma(\theta)\text{: its slice, net of costs}}$$

- **Revenue.** Spending on intermediates is a share η of output, and a fraction χ of that is markup, so profit before automation costs is **ηχ·Y**. The firm earns a cut of total output.
- **Cost.** Automation costs **C(θ)·Y**. Net profit is (ηχ − C(θ))·Y.
- **Trade-off.** More automation grows the pie, ln Y(θ) rises under (4), but shrinks the margin: Γ(θ) falls, faster and faster.

Key point: the monopolist maximizes *profit*, not output. It stops while ∂ln Y/∂θ is still positive, so a shock that moves θ has a **first-order** effect on Y (no envelope-theorem cancellation).

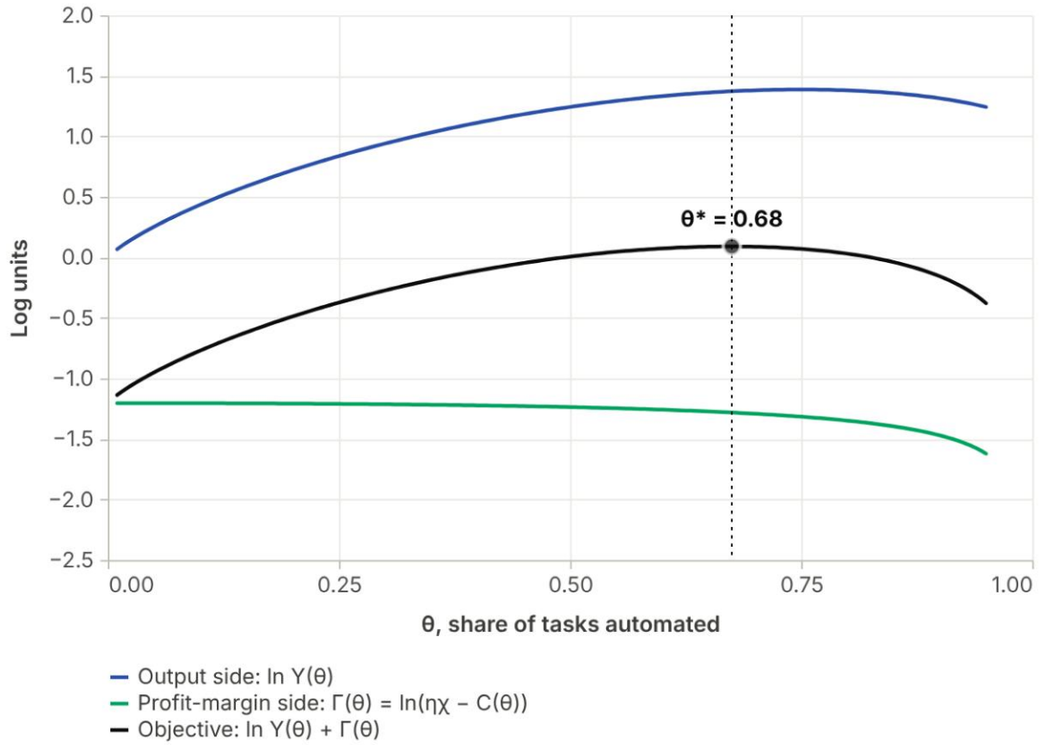


Illustration with K = 3, L = 1, ηχ = 0.3 and C(θ) = 0.05·(−θ − ln(1−θ)). χ is the Lerner index (p − 1)/p, which is why χ ∈ (0, 1); the paper calls it a markup. Hover for values.

Labor scarcity pushes automation up

First-order condition. Automate until the marginal gain in output equals the marginal loss in margin:

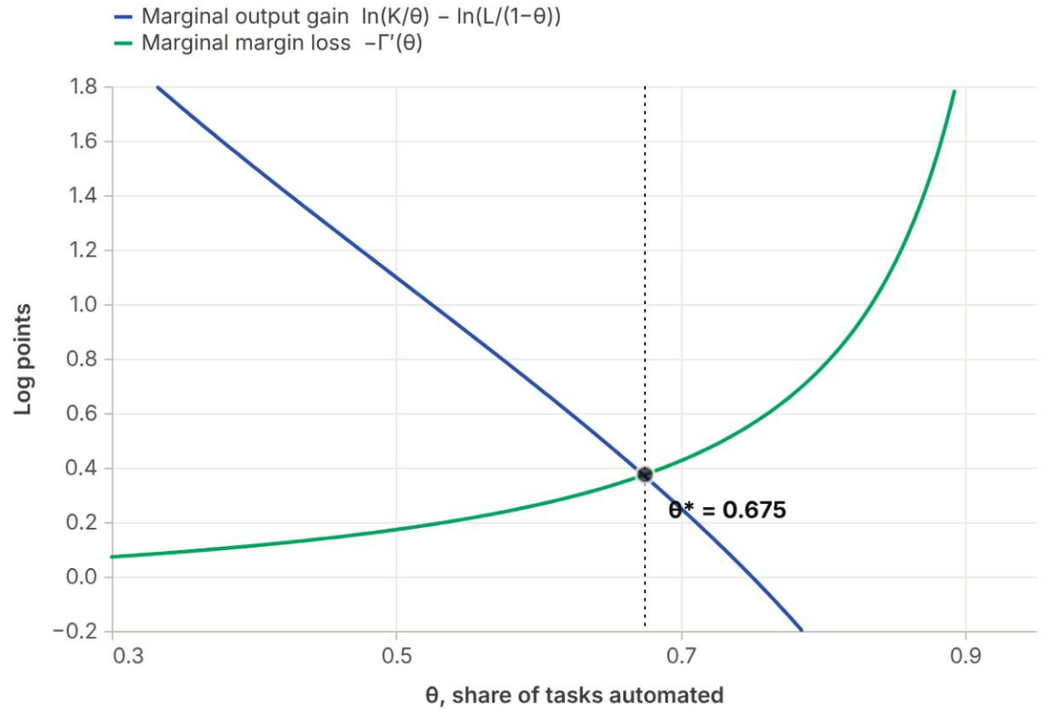
$$\ln \frac{K}{\theta} - \ln \frac{L}{1-\theta} \;=\; -\Gamma'(\theta)$$

Differentiate with respect to ln L:

$$\frac{d\theta}{d\ln L} = \frac{1}{\Gamma''(\theta) - \dfrac{1}{\theta(1-\theta)}} \;<\; 0$$

Aging prompts automation. A smaller young workforce (lower *L*) makes each manual task more labor-starved. The gap between capital per task and labor per task widens, which raises the payoff to automating, and θ* rises.

The sign needs only the output side: −1/(θ(1−θ)) < 0, and Γ″ < 0 only reinforces it.



Labor supply L (aging: drag left) 1

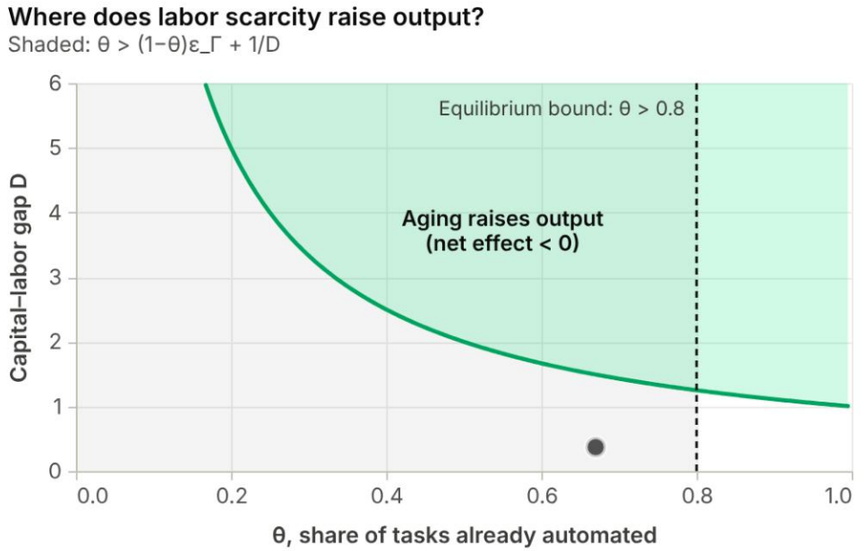
Same illustration as the previous slide (dashed: the gain curve at L = 1). Lowering L shifts the gain curve up, so it crosses the rising margin-loss curve further right.

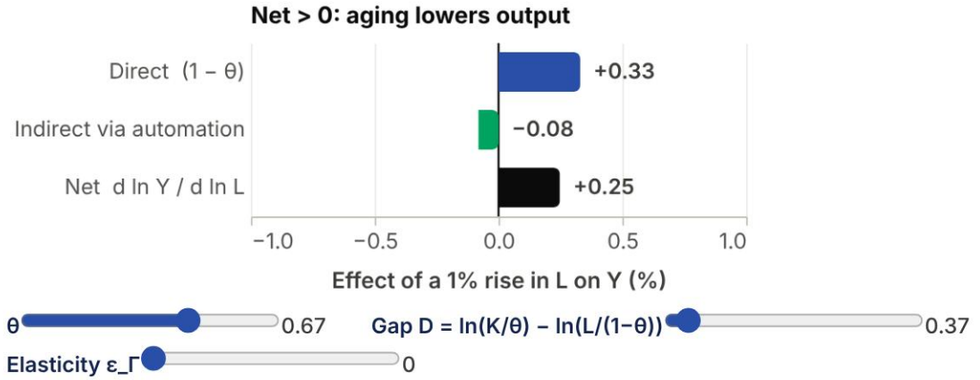
Direct vs. indirect effect of a smaller workforce

$$\frac{d\ln Y}{d\ln L} = \underbrace{1-\theta}_{\text{direct} > 0} - \underbrace{\frac{\theta(1-\theta)}{(1-\theta)\,\varepsilon_\Gamma + \dfrac{1}{D}}}_{\text{indirect, via automation} > 0}, \qquad D = \ln\tfrac{K}{\theta} - \ln\tfrac{L}{1-\theta}$$

- **Direct effect:** fewer workers means less output from manual tasks, so lower *L* lowers *Y*.
- **Indirect effect:** lower *L* raises θ (step 6), and each extra automated task adds *D* to ln *Y*.
- **Aging raises output when the indirect effect wins:** θ > (1 − θ)ε_Γ + 1/D. Ignoring the elasticity, this means **D > 1/θ**: a large capital–labor gap **and** substantial automation already in place.

Condition (4) is not enough. (4) only says D > 0; the reversal needs D > 1/θ. And ε_Γ is not free: at the monopolist's optimum, (1 − θ)ε_Γ + 1/D ≥ 2√(θ(1 − θ)), so a reversal requires **θ > 0.8**, whatever the cost function (our derivation). In the illustration on the last two slides, net = +0.27: aging lowers output.





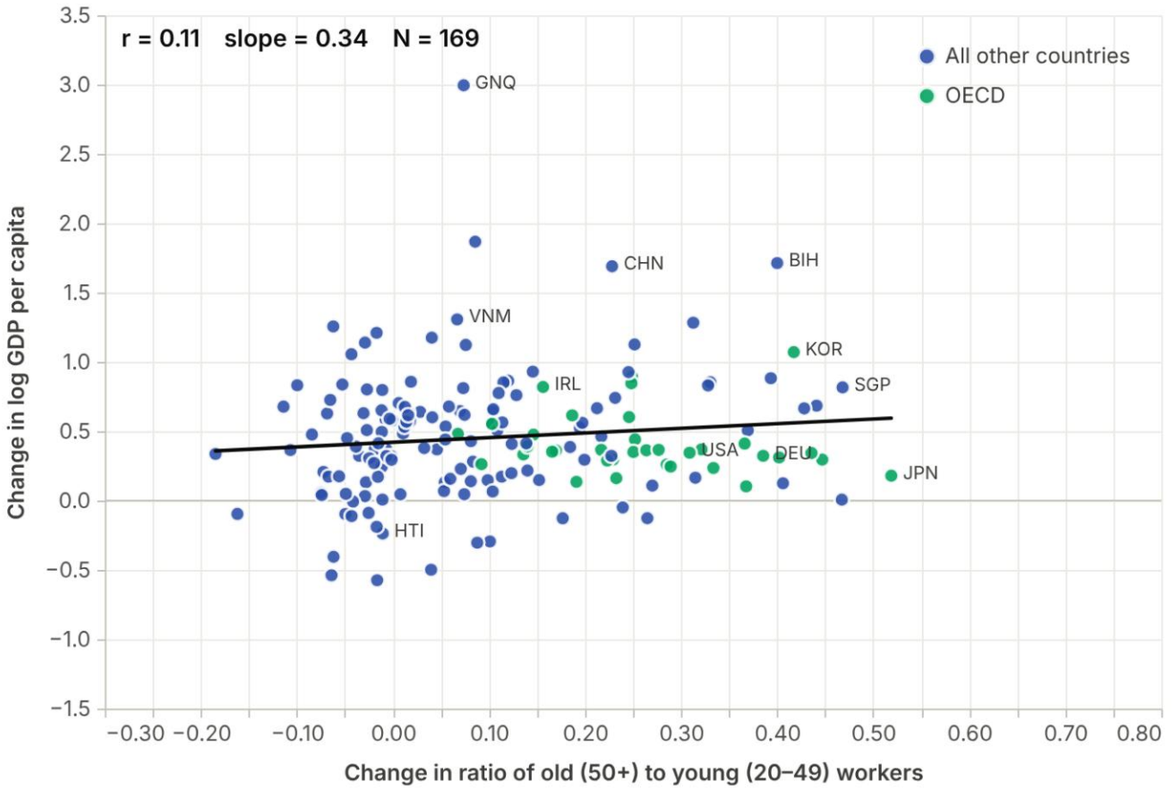
Faster-aging countries did not grow more slowly

The raw correlation is slightly positive, not negative. Countries that aged faster grew *a little faster*: r = 0.11, and the slope of 0.335 (s.e. 0.210) is not significant.

Consistent with the theory's punchline, a lower *L* need not lower *Y*, but note two gaps:

- The model is about **aggregate output** *Y*; the data are **GDP per capita**.
- The model is **static, in levels**; the data are **25-year growth rates**.

Equatorial Guinea (oil) is the outlier at the top. Dropping it changes little (column 4: 0.773 → 0.615).



Data ◉1990–2015, paper data ○1990–2023, PWT 11 + UN 2024

Figure 2 in the paper, rebuilt from the replication data. GDP: PWT 9.0, 1990–2014; aging: UN, 1990–2015. Switch to new data, hover for countries, click the legend to isolate a group. r and the fit line are recomputed in the browser.

INITIAL EVIDENCE II

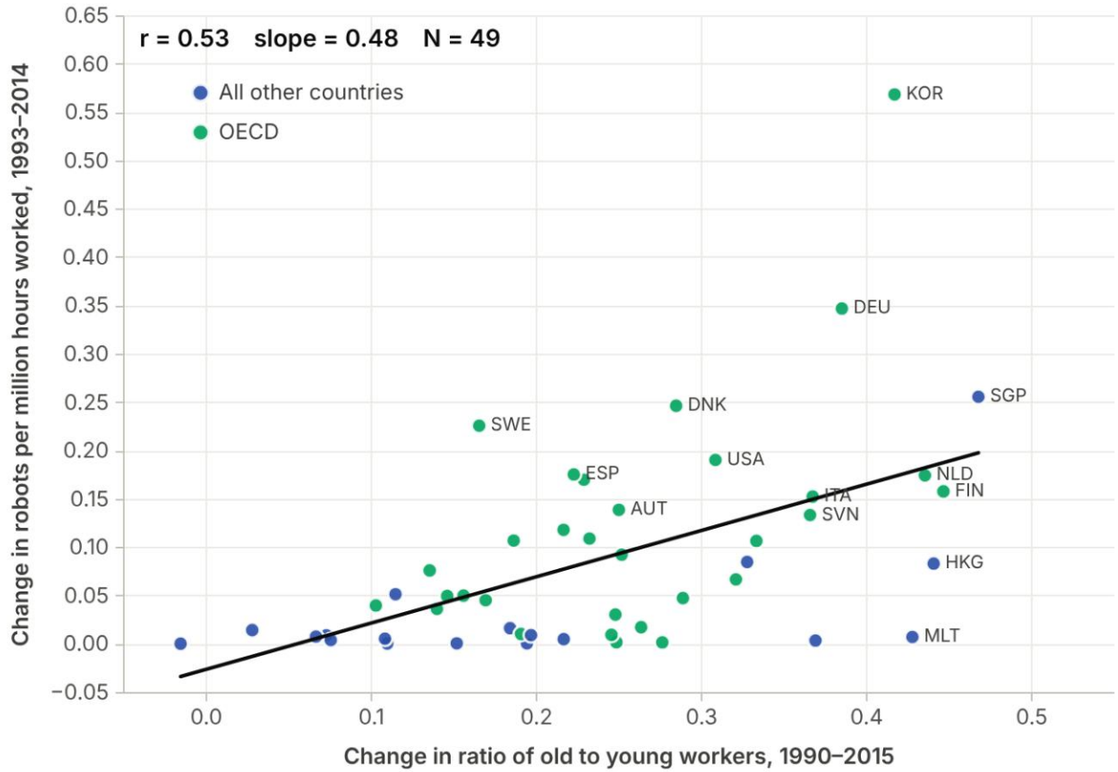
Aging goes hand in hand with robot adoption

Countries that aged faster adopted more industrial robots. r = 0.53 across 49 countries, and the relationship holds without Korea.

This is the **indirect effect** from step 6 showing up in the data: labor scarcity → automation.

> The authors concede this evidence is **"neither causal"** nor proof that robots are what neutralizes the effect of aging on growth.

Figure 3 in the paper (IFR data; Japan excluded by the IFR). The IFR data are proprietary, so the points were recovered from the vector paths of the published figure. The x-values match the replication data exactly (corr = 1.000), and OLS on the recovered points reproduces the paper's fit line.



Exclude Korea (the paper's outlier check) ☐

FROM CORRELATION TO EVIDENCE

Do these relationships survive stricter tests?

Two suggestive scatterplots are not evidence of a causal effect. The paper's regressions take on four alternative explanations, one at a time.

A raw correlation could be driven by…	…so the paper adds	Column
Convergence: poorer countries grow faster and are aging faster	Initial log GDP per capita (1990)	(2)
Starting point: young or large countries age mechanically faster	Initial old/young ratio and log population (1990)	(3)
Regional trends: East Asia aging *and* booming	Seven World Bank region dummies	(4)
Reverse causality: growth attracts young migrants and lowers mortality	IV: birthrates 1960–1990, which fix today's age structure before 1990	(5), (8)

Estimating equation (long difference, one row per country): $\mathbf{\Delta \ln(\text{GDP per capita})_{1990–2014} = \beta \cdot \Delta(\text{old/young})_{1990–2015} + \text{controls} + \varepsilon}$. Robust standard errors; columns 6–8 repeat the exercise for the 35 OECD countries.

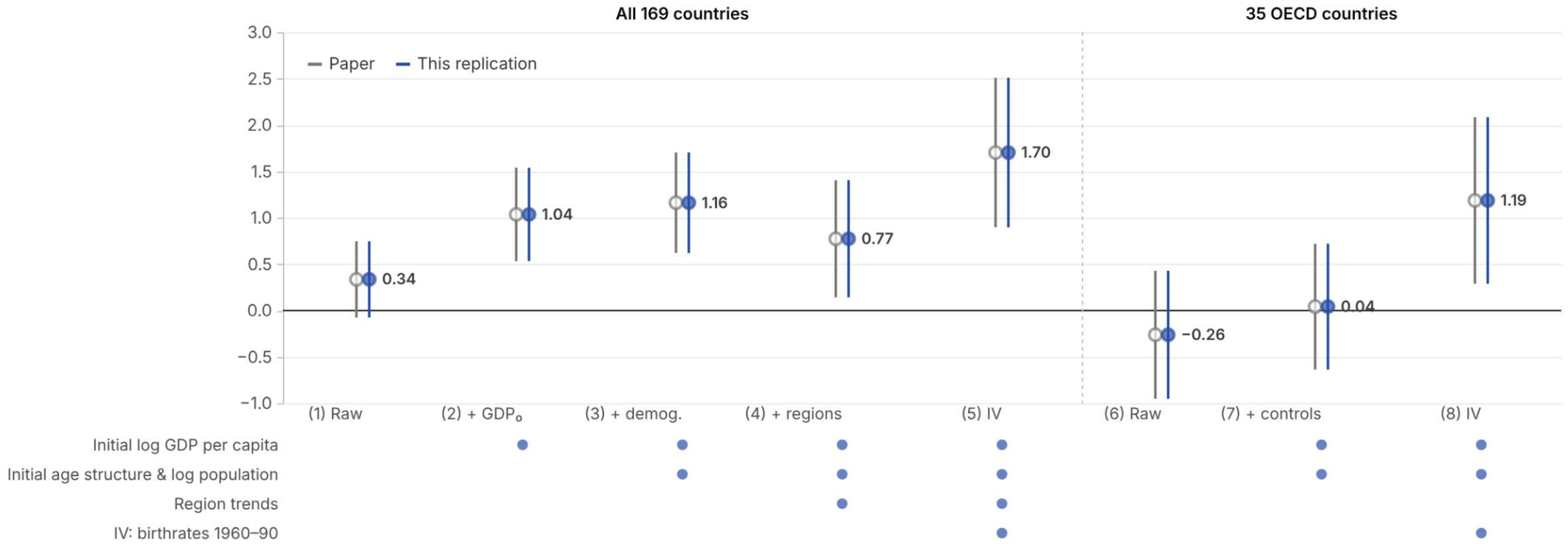
REPLICATION

Table 1: the positive relationship survives the controls

Columns add controls left to right. The full-sample estimate stays positive and significant; in the OECD only the IV is.

Cells matched to the paper	Col 4, all countries	Col 5, IV	Col 8, OECD IV
30 / 30	**0.77** (0.32)	**1.70** (0.41)	**1.19** (0.46)
β, s.e., F, J, N · Tables 1–2	with region trends	first-stage F = 19.4	first-stage F = 7.4 (weak)

Coefficient on aging with 95% confidence interval



Panel ◉A · old/young, 1990–2015 ○B · mean age 20+, 1990–2015 ○C · old/young, 2000–2015

Hollow gray: published estimate. Filled navy: this replication (Python, independent of Stata). The two sit side by side because they are identical. Switch panels to see the robustness checks. Coefficients in Panel B are per year of mean age.

REPLICATION

Stacked differences, and what replicating revealed

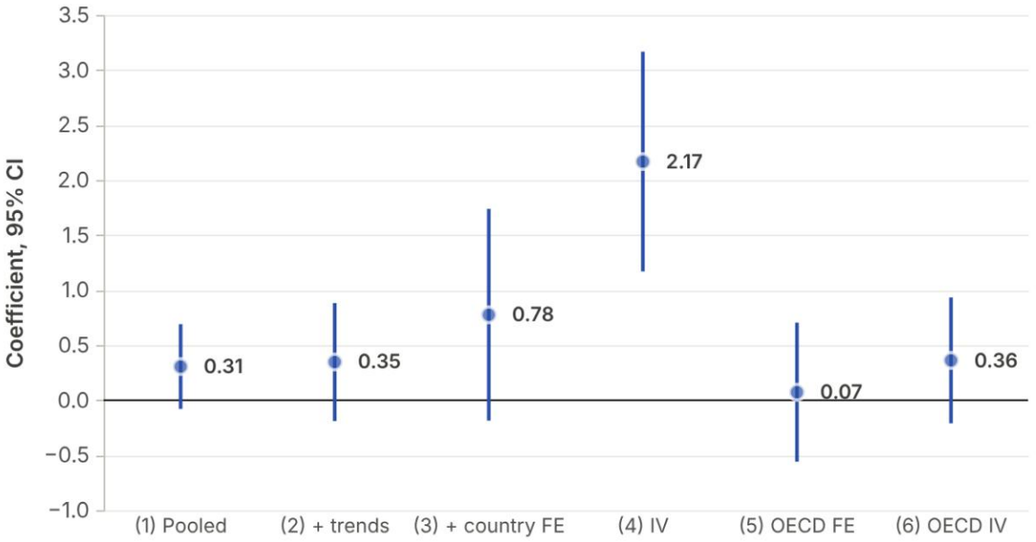
WHAT THE PAPER DOESN'T SAY (FOUND BY REPLICATING)

- **Six instruments, not five:** birthrates 1960–65 through **1985–90**. Only six reproduce F = 19.36.
- "Initial GDP **per worker**" is actually GDP **per capita**.
- GDP ends in **2014** (the last year of PWT 9.0); aging ends in **2015**.
- Table 2 lets only initial GDP have a period-specific slope. That is the only one of 16 variants that matches.

WEAK INSTRUMENTS

The OECD first stages are weak: **F = 7.4** (Table 1 col 8) and **F = 4.7** (Table 2 col 6), both below the rule-of-thumb 10. The paper does not discuss this.

Table 2: two stacked 25-year differences (1965–90, 1990–2015)
Country fixed effects absorb country-specific trends; SEs clustered by country



EXTENSION

Same question, new data, longer horizon

Data first (same 1990–2015 window), then longer windows, so we can tell whether a change comes from data revisions or from new years.

GDP per capita

~~Penn World Table 9.0, to 2014~~ → **PWT 11.0, to 2023**

2021 PPPs; China now uses its official series. Growth 1990–2014 correlates 0.93 with PWT 9.

Age structure & birthrates

~~UN WPP 2015~~ → **UN WPP 2024**

Old/young ratio, mean age of the 20+ population, 5-year birthrates rebuilt from UN data.

Windows

~~GDP 1990–2014, aging 1990–2015~~ → **both 1990–2019 and 1990–2023**

2019 avoids COVID; 2023 is the latest PWT year. The sample is held at the same 169 / 35 countries.

Instruments & inference

~~Birthrates 1960–90~~ → **also shifted forward, pre-1990 births only**

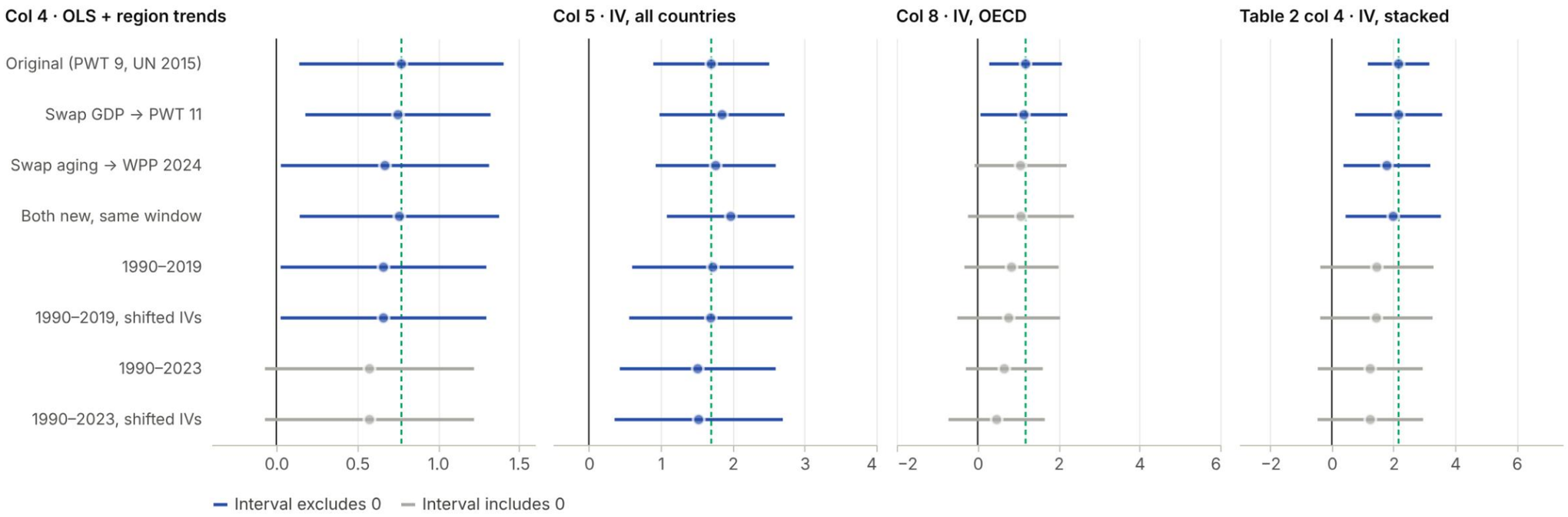
Added weak-instrument-robust Anderson–Rubin confidence sets, including a single-instrument version.

Why shifted instruments must stop at 1990. Shifting the windows forward keeps the link from birth cohort to the 20–49 age band, but births after 1990 can respond to growth over 1990–2023 itself. We keep only windows that end by 1990 (5 windows for 2019, 4 for 2023).

EXTENSION

The full-sample result holds; the OECD result does not

Each row is one run. Dashed green line: the original estimate. Gray: the interval includes zero.



Interval (●) Wald 95% CI (paper's method) ( ) Anderson–Rubin, 6 instruments ( ) Anderson–Rubin, 1 instrument (most robust)

Top four rows: same 1990–2015 window, data swapped one source at a time. Bottom four: longer windows, with original or shifted (pre-1990) instruments. The 6-instrument Anderson–Rubin set tests the coefficient and instrument validity jointly; the 1-instrument version (mean birthrate) tests the coefficient alone.

EXTENSION

What the extension does to the argument

Col 5 · IV, all countries
1.70 → 1.53
1990–2015 → 1990–2023 · still positive; borderline under the 1-instrument AR set

Col 4 · OLS + regions
0.77 → 0.58
erodes: part of the pattern is regional

Col 8 · IV, OECD
1.19 → 0.47
imprecise in every new-data run; 1-instrument estimate ≈ 1.0–1.5

Table 2 col 4 · stacked IV
2.17 → 1.25
intervals include zero once new years are added

WHAT IT MEANS FOR THE DIRECT AND INDIRECT EFFECTS

- **The regressions estimate only the net effect** (direct + indirect). A positive coefficient says that, on average, faster aging did not cost growth. In the model, that requires the indirect, automation channel to at least offset the direct loss of workers.
- **The full sample still says "no drag"** through 2023, so the stagnation prediction still fails.
- **But the model says the offset should be strongest where capital is abundant** (large *D*) and automation is advanced (high θ), which describes the OECD economies. That is exactly where the evidence **does not survive** new data.
- **COVID is not the story:** 2019 ≈ 2023.

Bottom line. The stylized fact (aging has not hurt growth) is robust. The evidence that *automation* is why it has not is much weaker than the paper suggests.

DISCUSSION

Strengths and weaknesses

STRENGTHS

- Takes on a popular thesis with a **clear, falsifiable fact**, and the fact replicates exactly.
- The full-sample result **survives** controls, IV, new data and nine more years.
- Offers a **mechanism** (directed technical change) with a testable prediction, not just a correlation.
- Short and transparent, with public data and code.

Your notes...

WEAKNESSES

- **Model ≠ regressions:** static, aggregate output in levels vs. 25-year growth in GDP *per capita*.
- **The reversal is a knife-edge:** it needs D > 1/θ, and in equilibrium **θ > 0.8**, i.e. most tasks already automated.
- **Weak OECD instruments** (F = 7.4, 4.7); birthrates may affect growth through other channels (demographic dividend, schooling).
- The robot evidence is a 49-country correlation the authors call "neither causal".
- **The OECD result doesn't survive** new data; the positive association sits mostly in **poorer** countries.
- Doesn't engage the monetary channel: Eggertsson, Lancastre & Summers (2019) find aging hurts growth at the zero lower bound.

Your notes...

THANK YOU

Questions?

Replication & extension code: Python (numpy, pandas, scipy); every number on these slides is validated against the paper.
Data: Acemoglu & Restrepo (2017) replication files · Penn World Table 11.0 · UN World Population Prospects 2024 · IFR via Figure 3.